



How to deploy production-grade AI entirely on-premises ?
zero bytes to the cloud, full regulatory compliance, lower lifetime cost.

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EXECUTIVE SUMMARY

The Sovereignty Problem

Why cloud AI fails the buyers who need it most

There is a quiet bifurcation happening in enterprise AI. On one side: the cloud AI vendors, racing to commoditize inference, dropping per-token prices, and offering "integration" that means one thing ? send us your data and we'll send back answers.

On the other side: a growing cohort of businesses that will never, under any circumstances, send their data to a third party. Not because they're paranoid. Because their regulators, their clients, and their own risk officers won't allow it.

These are law firms handling attorney-client privilege. Healthcare clinics bound by HIPAA. Financial institutions governed by PCI and GLBA. Engineering firms with proprietary IP. Family offices managing wealth for people who value discretion. They need AI. They cannot use the cloud. And almost nobody is building for them.

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THE CORE TENSION

The Local-First Stack

Five open-source tools that compose into production AI infrastructure

The technology to run production-grade AI entirely on-premises exists today and is mature. No vendor lock-in. No per-seat subscriptions. No data leaving your network. Here are the five components that compose into a sovereign AI stack:

OLLAMA Model Serving

Serves large language models locally ? the same model families the cloud vendors offer (GLM, Kimi, DeepSeek, Llama, Mistral), running on hardware you own, inside your network, with zero bytes transmitted externally.

CHROMADB Vector Storage

Provides semantic search across your document corpus. Embeddings never leave your building. Enables retrieval-augmented generation so the model answers from your documents, not from training data hallucinations.

DOCKER Containerization

Wraps the entire stack in reproducible, upgradable, disaster-recoverable containers. No snowflake servers. No dependency hell. Backups and restores are trivial. The entire AI infrastructure is version-controlled infrastructure-as-code.

N8N Workflow Orchestration

Connects the AI layer to your existing tools ? document intake, classification, routing, compliance logging ? without exposing data to external APIs. Every workflow is visible, auditable, and modifiable by your team.

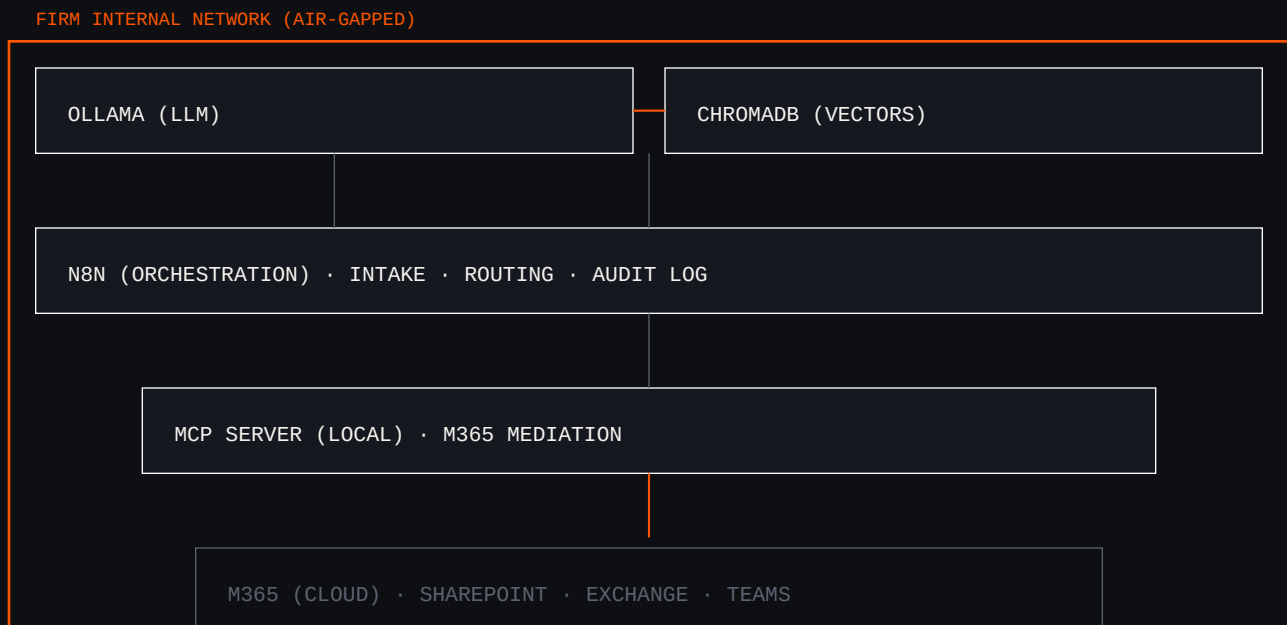
MCP Protocol Mediation

Model Context Protocol provides a standardized interface between the AI infrastructure and cloud productivity suites like M365. Your SharePoint, Exchange, and Teams data is mediated by a local server that controls what the AI can see and do ? on your terms.

Architecture Blueprint

A production reference diagram ? no confidentiality required

The architecture below runs in production today. It is drawn without any client-specific detail ? every component is open-source and deployable by any qualified engineering team. The diagram shows the fundamental principle: all AI compute and data storage stays inside the firm's network boundary. Only structured, mediated API calls reach external services.



- Sovereign compute · LLM runs on hardware you own, inside your network. No external API calls.
- Local vector storage · embeddings never leave the building. Your corpus stays your corpus.
- MCP mediation · cloud integrations are brokered by a local server you control.
- Full audit trail · every interaction logged. Demonstrate compliance without exposing data.
- Docker-containerized · upgrades, backups, and disaster recovery are trivial and scripted.

Security: The Fortress Protocol

Zero-trust hardening before any AI goes in

Deploying AI on a vulnerable network creates new attack surfaces. The security posture must be assessed and hardened before any model is loaded. This is not optional ? it is professional malpractice to deploy AI infrastructure on a network you haven't audited.

PHASE 1 Network Topology Mapping

Identify every data flow path for regulated information, from intake to archival. Document what goes where, through what protocol, authenticated by what mechanism. You cannot secure what you have not mapped.

PHASE 2 Access Control Audit

Review Active Directory group policies, file share permissions, and remote access configurations against zero-trust principles. The most common finding: legacy VPN configs that grant broader access than any current role requires.

PHASE 3 Gap Analysis

Document where current practices fall short of regulatory obligations ? privilege standards, records retention, breach notification protocols. Prioritize gaps by risk severity and remediation cost.

PHASE 4 AI-Readiness Assessment

Classify which workflows are safe for AI augmentation versus which require human-only handling. Not every document should touch a model. The assessment defines the boundary.

FORTRESS PRINCIPLE

Cost Analysis: Sovereign vs Cloud

The three-year economics for a 60-person regulated firm

Local-first AI is not just a compliance play. It is an economic one. The cost crossover happens in year two ? and the gap widens every year after as you stop paying per-seat for inference that runs on hardware you already own.

	CLOUD SaaS	SOVEREIGN
Year 1 · Hardware	Included	\$12K (one-time)
Year 1 · Deployment	\$0 (self-serve)	\$35K (one-time)
Year 1 · Subscription/Retainer	\$144K (\$200/seat × 60)	\$54K (\$4.5K/mo)
Year 1 · Legal/BAA Review	\$20K (one-time)	\$0
Year 1 TOTAL	\$164K	\$101K
Year 2 · Subscription/Retainer	\$144K	\$54K
Year 3 · Subscription/Retainer	\$144K	\$54K
3-YEAR TOTAL	\$452K	\$209K

THE SOVEREIGNTY PREMIUM

Deployment Roadmap

A 6-week implementation plan from audit to production

WEEK 1-2 Fortress Security Audit

- Network topology mapping
- Access control audit (AD, file shares, VPN)
- Gap analysis against regulatory obligations
- AI-readiness classification of workflows
- Deliverable: 40-page security posture report

WEEK 3-4 Infrastructure Deployment

- Provision GPU workstation on internal network
- Deploy Ollama with locally-quantized model
- Deploy ChromaDB + index document corpus
- Deploy n8n orchestration layer
- Deploy MCP server for M365 mediation

WEEK 5 Integration + Workflows

- Connect AI stack to existing M365 environment
- Build document intake automation workflow
- Build template drafting workflow
- Configure Slack/Teams integration
- Enable full audit logging

WEEK 6 Validation + Handoff

- End-to-end workflow testing with real documents
- Compliance officer review of audit trail
- Team training (pull-based, not mandatory sessions)
- Documentation handoff (architecture + runbooks)
- Transition to managed retainer model

The Manteis Engagement Model

Audit ? Deploy ? Manage ? an integrated lifecycle, not a product

Most AI consultancies sell one of two things: a SaaS subscription ("send us your data") or a one-time deployment ("we'll set up Ollama for you"). Neither works for regulated industries. The first violates compliance. The second leaves the firm with infrastructure that decays ? stale models, stagnant workflows, no security monitoring.

Manteis sells a third thing: the integrated lifecycle. Security audit first, because deploying AI on a vulnerable network is malpractice. Sovereign infrastructure second, because the technology exists and the architecture is proven. Managed retainer third, because AI is not a set-and-forget tool ? it is infrastructure that needs tending.

01 FORTRESS AUDIT

\$12K

2 weeks

Full security assessment of existing data handling. Network topology, access controls, gap analysis, AI-readiness classification. The audit pays for itself ? it frequently finds pre-existing vulnerabilities that predate the AI engagement.

02 INFRASTRUCTURE DEPLOY

\$35K

6 weeks

Sovereign AI stack deployed: Ollama, ChromaDB, Docker, n8n, MCP. Integrated with existing M365 environment. Production workflows live. Full audit trail. Zero bytes of client data leave the network. The firm owns the infrastructure.

03 MANAGED RETAINER

\$4.5K/mo

Ongoing

Model updates, workflow optimization, security monitoring, use-case development. Quarterly workshops to identify new AI applications. The infrastructure is maintained, not abandoned. AI is a garden, not a server you rack and forget.

WHAT YOU BUY

Decision Framework

Should your organization go sovereign?

? Does your compliance officer reject cloud AI vendors?

If yes, sovereign AI isn't an option ? it's the only option.

? Is your data regulated, privileged, or competitively sensitive?

Regulated (HIPAA, PCI, GLBA), privileged (attorney-client), or proprietary (trade secrets) all point the same direction: on-premises.

? Do you have 60+ knowledge workers doing repetitive document work?

Document intake, classification, template drafting, precedent retrieval ? these are the workflows where local AI delivers 70-90% time reduction.

? Can you provision a GPU workstation on your internal network?

A single workstation (\$8-15K) serves a 60-person firm. No data center required. Air-gapped from internet, accessible via existing VPN.

? Do you need a competitive moat the SaaS model can't provide?

The sovereign firm customizes models, tunes workflows, and builds proprietary automation. The SaaS customer rents a black box. The moat is ownership.

NEXT STEP

Book a free 30-minute Sovereignty Audit.